

Sister, Sister: Ependymal Cells and Adult Neural Stem Cells Are Separated at Birth by Geminin Family Members

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The adult subventricular zone (SVZ) stem cell niche is comprised of multi-ciliated ependymal cells that line the brain ventricular system and adult stem cells. Papers in *Neuron* (Ortiz-Álvarez et al., 2019) and *Cell Reports* (Redmond et al., 2019) report that these cell types share a common precursor. Ortiz-Álvarez et al. further show that Geminin family members modulate the fate of daughter cells.

The central nervous system (CNS) is famously unable to regenerate the vast majority of neuronal cell types. Re-stimulating this ability in the context of degeneration or disease is the holy grail of CNS regenerative medicine, but this active field of research is fraught with quiescent, lineage-restricted stem cells, the need for complex re-wiring, and a dearth of survival factors present at earlier developmental stages. However, adult neuronal stem cells do exist and provide tantalizing prospects for CNS regeneration. In the mouse, stem cell pools generate both hippocampal neurons in the subgranular zone (SGZ) of the dentate gyrus and olfactory bulb interneurons (mostly granule neurons) from the subventricular zone (SVZ) adjacent to the lateral ventricles. Although these progenitors are maintained into adulthood, their potential and ability to proliferate decreases with age. The unique stem cell pools in the SVZ require a precisely formed, well-controlled niche that includes maintaining contact with the cerebrospinal fluid (CSF) and blood supply (Bjornsson et al., 2015; Silva-Vargas et al., 2016). Together, the studies by Ortiz-Álvarez et al. in *Neuron* (Ortiz-Álvarez et al., 2019) and Redmond et al. in *Cell Reports* (Redmond et al., 2019) shed light on the delicate balance required to establish the adult SVZ during much earlier stages of embryogenesis.

Ependymal cells lining the walls of brain ventricles play essential roles in CSF flow and in brain homeostasis (Spassky et al., 2005). Ependymal cells (E1 cells) are distinct from both radial-glial-like adult

stem cells (B1 cells) and multi-ciliated, choroid-plexus epithelial cells, although all three cell types directly contact the CSF. This “lakefront property” also contains transient amplifying cells (C cells), neuroblasts (A cells), astrocytes that arise from B1 cells, and vasculature. Although radial-glial progenitors can give rise to both ependymal cells and adult neural stem cells (Spassky et al., 2005), the direct lineage for these two crucial cell types has remained elusive.

Álvarez, Redmond, and colleagues now provide new pieces to this puzzle and begin to unlock the mystery shrouding the ontogeny of the adult SVZ niche. Detailed histology of the mouse-brain ventricular surface, including the apical surface area per cell and immunofluorescence for VCAM1 (B1 cells) and Foxj1 (multi-ciliated cells), allowed Redmond, et al. (Redmond et al., 2019) to determine that E1 and B1 cells are fully distinguishable from each other by postnatal day 5 (P5) in mice, and the cells begin to arrange in exquisitely organized pinwheels at this time, as well (Redmond et al., 2019). This differentiation program is highly regionalized and stereotyped, beginning posteriorly and ventrally and proceeding anteriorly and dorsally. By using Cre-driven lineage tracing, Ortiz-Álvarez et al. (Ortiz-Álvarez et al., 2019) determined that ependymal cells are produced locally from ventricular radial glia of the *Emx1*, *Gsh2*, and *Nkx2.1* lineages. By using state-of-the-art lineage tracing constructs—either NucBow/MAGIC Marker (Loulrier et al., 2014) or UbC-StarTrack

(Figueroes-Oñate et al., 2016) delivered by *in utero* electroporation—paired with thymidine analog birthdating, both groups determined that the division of an individual radial-glial cell could result in E1-B1 mixed clones (Ortiz-Álvarez et al., 2019; Redmond et al., 2019) in addition to E1-E1 clones. Through their analysis of differentiated neurons in the olfactory bulb, Ortiz-Álvarez et al. (Ortiz-Álvarez et al., 2019) also identified B1-B1 clones. From these data, both groups used independent experimental paradigms to show that ependymal cells and adult neural stem cells share a common embryonic precursor.

Because E1 cells are postmitotic and are derived between embryonic day 13.5 (E13.5) and E15.5 in mice (Spassky et al., 2005), both groups asked whether the proportion of E1-only or B1-E1 mixed clones changed over developmental time. Indeed, clones were larger at E13.5, mixed clones peaked earlier (E13.5; Ortiz-Álvarez et al., 2019), and E1-only clones peaked later (E14.5; Ortiz-Álvarez et al., 2019). These studies indicate that the potential of a radial-glial progenitor to produce mixed E1-B1 clones shifts as neural development proceeds, consistent with the potential of the radial-glial lineage to contribute to an array of cell types for the developing brain.

But what controls this lineage restriction switch? Ortiz-Álvarez et al. hypothesized that the antagonist regulators of DNA replication, GemC1 and Geminin, could tune this process because Geminin superfamily members have previously

been shown to regulate multi-ciliogenesis (Arbi et al., 2016; Terré et al., 2016). By combining mosaic analysis with double markers (MADM) of clones and GemC1 mis-expression, Ortiz-Álvarez et al. (Ortiz-Álvarez et al., 2019) induced premature ependymal cell differentiation at the expense of B1 adult neural progenitor cells. No B1-B1 clones were observed after GemC1 expression (Ortiz-Álvarez et al., 2019). The GemC1 antagonist Geminin was recently shown to be expressed by cycling type B1 cells in the adult SVZ (Sankar et al., 2016), and hence, its roles in radial-glia cells and newborn E1 progenitors are of key interest to this differentiation paradigm. By combining MADM of clones and Geminin overexpression, the authors could induce type B1-B1 clones that generate adult neural progenitor cells in excess, although the overall number of E1-B1 mixed clones was unchanged by this insult, suggesting additional regulatory mechanisms (Ortiz-Álvarez et al., 2019). Therefore, although GemC1 is sufficient to shift the balance toward E1 cells, and Geminin can shift the population toward B1 cells, the requirement for these two gene products in the differentiation processes of E1 and B1 cells remains an open question.

These studies provide complementary evidence that a radial-glia progenitor can produce two components of the adult stem cell niche with vastly different properties and functions. These are critical to brain function and are generated by all of the major radial-glia lineages, including

those specified to be pallial (Emx1) or subpallial (Gsh2-LGE and Nkx2.1-MGE). The involvement of Geminin superfamily members suggests that the drivers of this fate decision include not only asymmetric inheritance of cellular components such as centrioles or mitochondria but also, potentially, DNA-replication events. Instructive signals delivered by the adjacent CSF could also provide key ingredients for shaping this fundamental process. Indeed, the regionalized organization and maturation course of the E1 cells raises provocative questions regarding the formation of early gradients of CSF-borne factors throughout the embryonic ventricular system. Because the adult SVZ feeds the olfactory bulb with new neurons via the rostral migratory stream, collectively these studies inspire new hypotheses regarding how mechanisms involved in establishing the blueprint for this multi-cellular network at such early developmental ages influences later olfaction and the regenerative and adaptive capacity of the SVZ-olfactory bulb axis throughout life.

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